

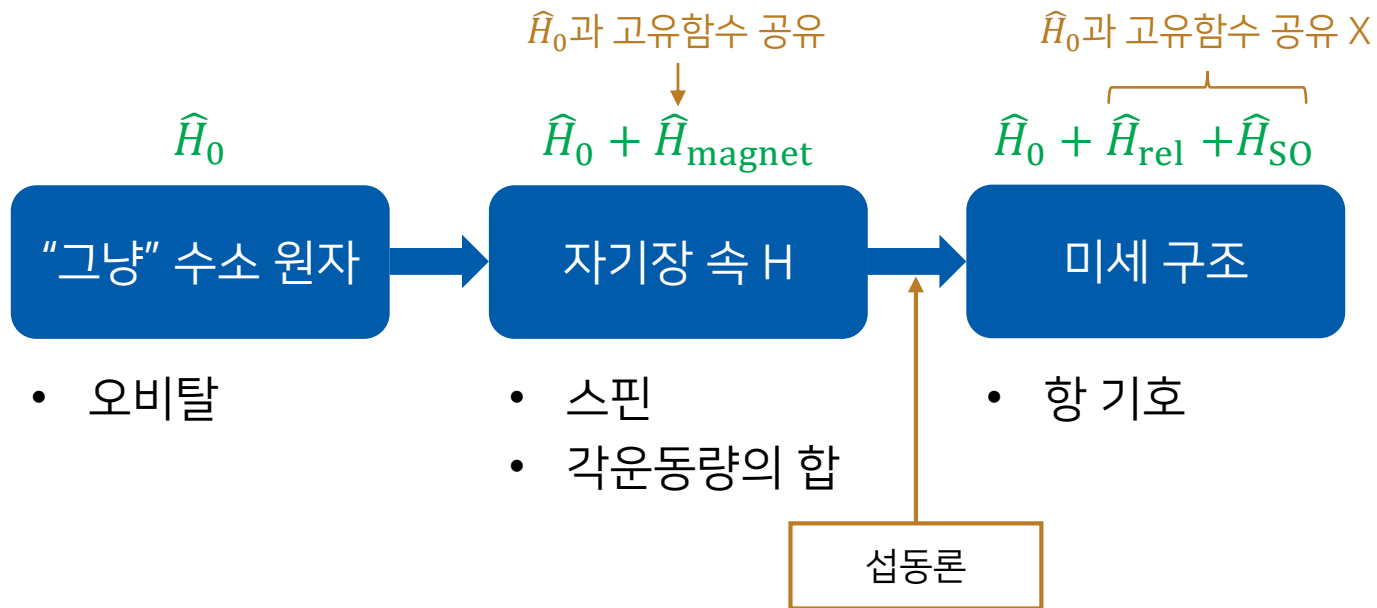
Physical Chemistry (II)

Lecture 13

Fine Structure of Hydrogen



Where Are We: Hydrogen Atom



Last class

Full Description of H-Atom Electron

- Quantum numbers: n, l, m_l, m_s (s is fixed)
- Wavefunction:

$$\psi_{nlm_l}(r, \theta, \phi) |m_s\rangle = |n \ l \ m_l \ m_s\rangle$$

- Energy (no magnetic field):

$$E_n = -\frac{ZR_H}{n^2}$$

- Energy (in a magnetic field):

$$E = -\frac{ZR_H}{n^2} + ?$$

Last class

In a Strong Field

Two magnetic moments independently react to the field

$$E_{\text{magnet}} = E_{\text{orbital}} + E_{\text{spin}}$$

$$E_{\text{orbital}} = \beta_B m_l B$$

$$E_{\text{spin}} = g\beta_B m_s B$$

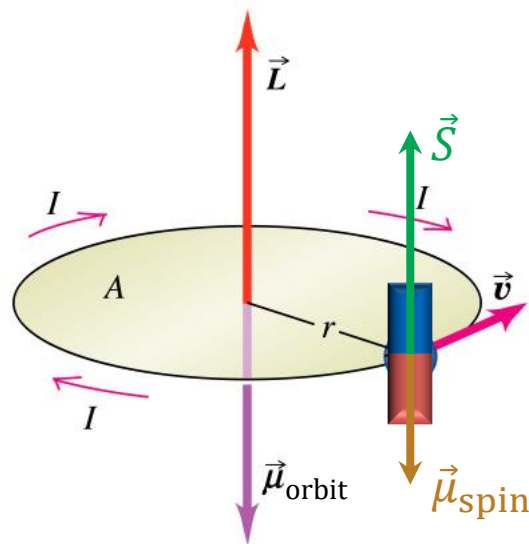
$$E = E_n + E_{\text{magnet}} = -Z \frac{R_H}{n^2} + \beta_B B (m_l + 2m_s)$$



Last class

In a Weak Field

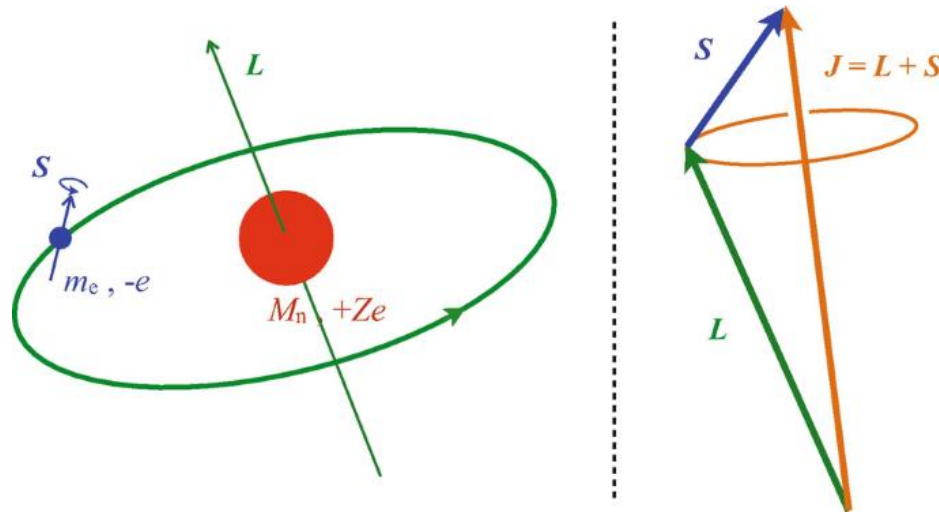
Need to consider the **spin-orbit coupling**



How?

Total Angular Momentum

- Orbital and spin angular momenta can be added
- Angular momenta are vectors ($\vec{L}$ and $\vec{S}$)
- The total angular momentum $\vec{J} = \vec{L} + \vec{S}$ is conserved



(Image: Kastberg, "The Spin-Orbit Interaction", in *Structure of Multielectron Atoms*, 2020)

Last class

QM of Total Angular Momentum

- Eigenfunctions $|j\ m_j\rangle$
- Eigenvalues

$$\hat{J}^2 |j\ m_j\rangle = j(j+1)\hbar^2 |j\ m_j\rangle$$

$$\hat{J}_z |j\ m_j\rangle = m_j\hbar |j\ m_j\rangle$$

- Quantum numbers
 - Allowed $m_j = -j, -(j-1), \dots, (j-1), j$
 - Allowed $j = (l+s), (l+s-1), \dots, |l-s|$

Weak-Field Hamiltonian

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 - \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r} + \frac{e}{2\mu} (\vec{L} + 2\vec{S}) \cdot \vec{B}$$

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 - \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r} + \frac{e}{2\mu} \left[1 + \frac{1}{2\hat{j}^2} (\hat{j}^2 + \hat{s}^2 - \hat{l}^2) \right] \vec{j} \cdot \vec{B}$$

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 - \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r} + \frac{eB}{2\mu} \left[1 + \frac{1}{2\hat{j}^2} (\hat{j}^2 + \hat{s}^2 - \hat{l}^2) \right] \hat{j}_z$$

$$E = -Z \frac{R_H}{n^2} + \beta_B B \left(1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)} \right) m_j$$

Landé g factor $g(j, l)$

Last class

"Good" Quantum Numbers

Two sets of quantum numbers: (n, l, m_l, m_s) and (n, l, j, m_j)

- Strong $\vec{B}$: (n, l, m_l, m_s) describes the state better
- Weak $\vec{B}$: (n, l, j, m_j) describes the state better

Perturbation Theory

$$\hat{H}\psi = E\psi$$

We can apply the perturbation theory if the Hamiltonian can be decomposed as

$$\hat{H} = \hat{H}_0 + \hat{H}'$$

unperturbed Hamiltonian
↓
↑
perturbation

where we know $E_n^{(0)}$ and $\psi_n^{(0)}$ (n : quantum number):

$$\hat{H}_0\psi_n^{(0)} = E_n^{(0)}\psi_n^{(0)}$$

- Note: $\hat{H}_0$ and $\hat{H}'$ do not share the eigenfunctions

Series Expansion

Introduce the bookkeeping parameter, λ :

$$\hat{H} = \hat{H}_0 + \lambda \hat{H}'$$

and assume that

$$\psi_n = \psi_n^{(0)} + \lambda \psi_n^{(1)} + \lambda^2 \psi_n^{(2)} + \dots$$

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots$$

Substitute everything into $\hat{H}\psi_n = E_n\psi_n$ and collect terms...

Different Orders

Zeroth order (λ^0):

$$\hat{H}_0 \psi_n^{(0)} = E_n^{(0)} \psi_n^{(0)} \text{ (what we knew)}$$

First order (λ^1):

$$\hat{H}_0 \psi_n^{(1)} + \hat{H}' \psi_n^{(0)} = E_n^{(0)} \psi_n^{(1)} + E_n^{(1)} \psi_n^{(0)}$$

Second order (λ^2):

$$\hat{H}_0 \psi_n^{(2)} + \hat{H}' \psi_n^{(1)} = E_n^{(0)} \psi_n^{(2)} + E_n^{(1)} \psi_n^{(1)} + E_n^{(2)} \psi_n^{(0)}$$

$\vdots$

Last class

First-Order Energy

$$E_n^{(1)} = \int \psi_n^{(0)*} \hat{H}' \psi_n^{(0)} d\tau$$

Dirac Bracket Notation

- Ket: $|n\rangle = \psi_n(x)$
- Bra: $\langle m| = \int \psi_m^*(x) \dots dx$
- Bra-ket

$$\langle m|n\rangle = \int \psi_m^*(x) \psi_n(x) dx$$

- With an operator

$$\langle m|\hat{A}|n\rangle = \int \psi_m^*(x) \hat{A} \psi_n(x) dx$$

Dirac Bracket Notation

- Normalization and orthogonality: for $m \neq n$,

$$\langle n|n\rangle = \int \psi_n^*(x)\psi_n(x) dx = 1$$

$$\langle m|n\rangle = \int \psi_m^*(x)\psi_n(x) dx = 0$$

- Expectation value

$$\langle n|\hat{A}|n\rangle = \int \psi_n^*(x)\hat{A}\psi_n(x) dx$$

First-Order Energy

- Integral notation

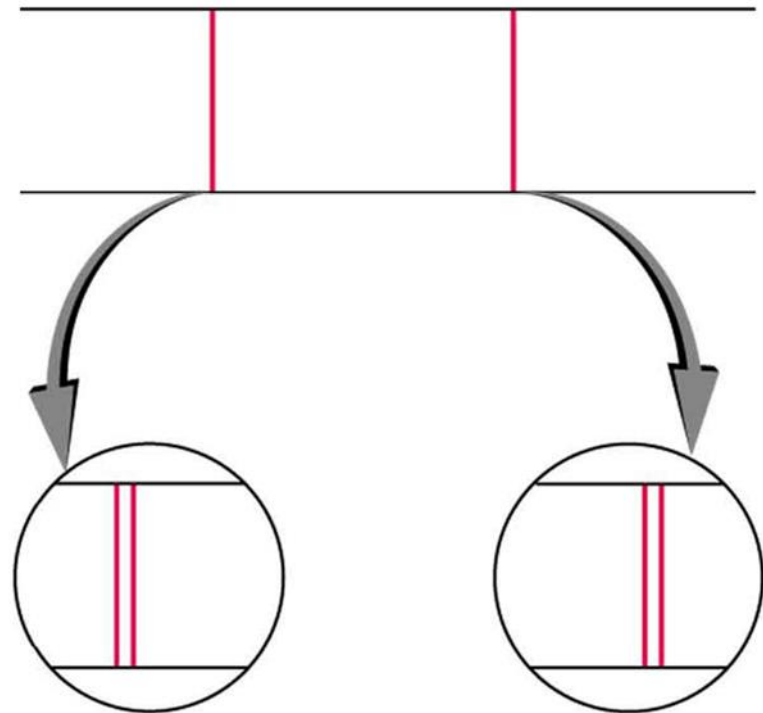
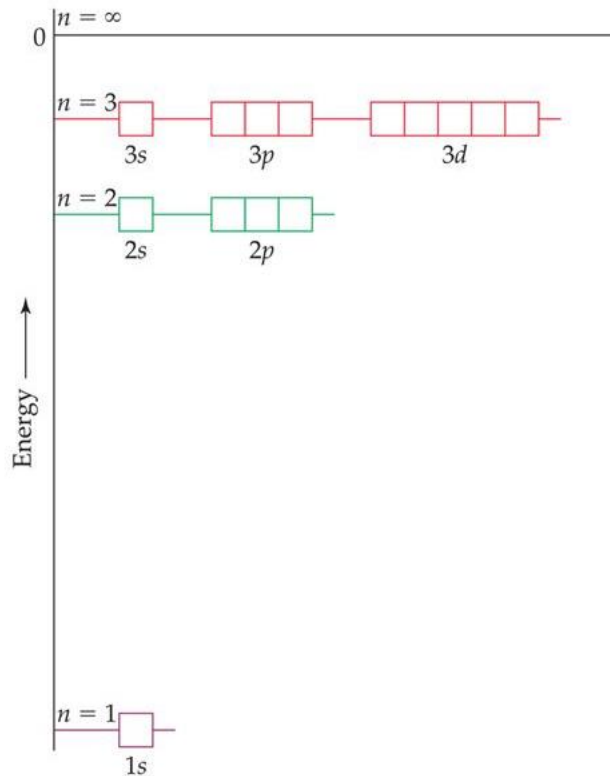
$$E_n^{(1)} = \int \psi_n^{(0)*} \hat{H}' \psi_n^{(0)} d\tau$$

- Bracket notation

$$E_n^{(1)} = \left\langle \psi_n^{(0)} \left| \hat{H}' \right| \psi_n^{(0)} \right\rangle$$

Last class

Fine Structure of Hydrogen



Origin of Fine Structure

1. Relativistic effect
2. Spin-orbit coupling

$$\hat{H}_0 \rightarrow \hat{H}_0 + \hat{H}_{\text{rel}} + \hat{H}_{\text{so}}$$

$\hat{H}_{\text{rel}}$ or $\hat{H}_{\text{so}}$ do NOT share eigenfunctions with $\hat{H}_0$!

(Note: the two effects are relatively small)

Relativistic Effect

- Kinetic energy in CM

$$T = \frac{p^2}{2m}$$

- Kinetic energy in relativity

$$T = \sqrt{p^2 c^2 + m^2 c^4} - mc^2 = \frac{p^2}{2m} - \frac{p^4}{8m^3 c^2} + \dots$$

$\hat{H}_{\text{rel}}$

- QM: first-order energy from perturbation theory

$$E_{\text{rel}} = \langle n \ l \ m_l \ m_s | \left(-\frac{\hat{p}^4}{8m^3 c^2} \right) | n \ l \ m_l \ m_s \rangle$$

Relativistic Correction

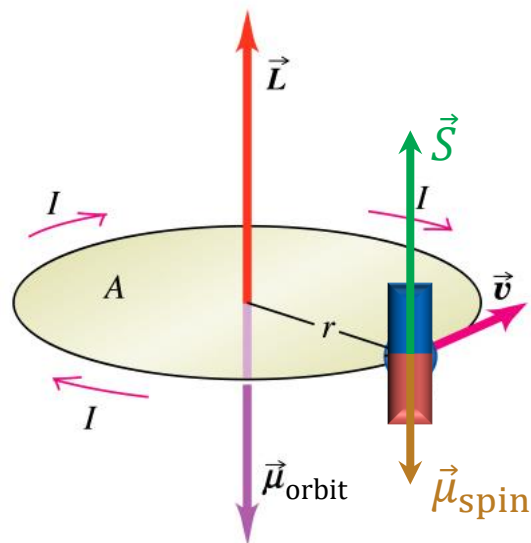
$$\begin{aligned} E_{\text{rel}} &= -\frac{1}{8m^3c^2} \langle n \ l \ m_l \ m_s | \hat{p}^4 | n \ l \ m_l \ m_s \rangle \\ &= -\frac{1}{8m^3c^2} \langle m_s | m_s \rangle \int \psi_{nlm_l}^* \hat{p}^4 \psi_{nlm_l} dV \end{aligned}$$

After tedious mathematics, one obtains

$$E_{\text{rel}} = -\frac{(E_n)^2}{2mc^2} \left(\frac{4n}{l + 1/2} - 3 \right)$$

M 7.6

Spin-Orbit Coupling (in No Field)



$\vec{\mu}_{\text{spin}}$ in a magnetic field generated by $\vec{\mu}_{\text{orbit}}$

Magnetic Field

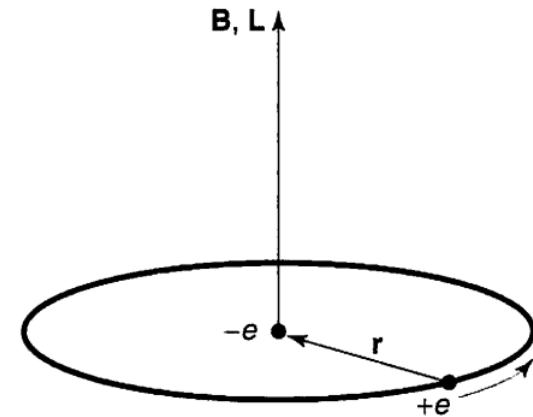
- From electron's perspective
- Biot-Savart law

$$B = \frac{I}{2\epsilon_0 c^2 r}$$

Using the current $I = ev/(2\pi r)$,

$$B = \frac{1}{4\pi\epsilon_0} \frac{ev}{c^2 r^2} = \frac{1}{4\pi\epsilon_0} \frac{e}{mc^2 r^3} L$$

$$\vec{B} = \frac{1}{4\pi\epsilon_0} \frac{e}{mc^2 r^3} \vec{L}$$



Spin in a Magnetic Field

$$\vec{B} = \frac{1}{4\pi\epsilon_0} \frac{e}{mc^2 r^3} \vec{L}$$

- Potential energy $V = -\vec{\mu}_{\text{spin}} \cdot \vec{B}$
- Magnetic moment

$$\mu_{\text{spin}} = -\frac{eg}{2m} \vec{S} \approx -\frac{e}{m} \vec{S}$$

$$V = \left(\frac{e}{m} \vec{S} \right) \cdot \left(\frac{1}{4\pi\epsilon_0} \frac{e}{mc^2 r^3} \vec{L} \right) = \frac{e^2}{4\pi\epsilon_0} \frac{1}{m^2 c^2 r^3} \vec{S} \cdot \vec{L}$$

Thomas Precession

Relativistic effect on accelerating particle

... which adds one more correction:

$$V = \frac{e^2}{4\pi\epsilon_0} \frac{1}{m^2 c^2 r^3} \vec{S} \cdot \vec{L} \rightarrow \frac{e^2}{8\pi\epsilon_0} \frac{1}{m^2 c^2 r^3} \vec{S} \cdot \vec{L}$$

Potential energy due to spin-orbit coupling!

Write in Quantum Mechanics

$$V = \frac{e^2}{8\pi\epsilon_0} \frac{1}{m^2 c^2 r^3} \vec{S} \cdot \vec{L}$$

Using $\vec{L} \cdot \vec{S} = \vec{S} \cdot \vec{L} = (J^2 - L^2 - S^2)/2$,

$$V = \frac{e^2}{8\pi\epsilon_0} \frac{1}{m^2 c^2 r^3} \frac{J^2 - L^2 - S^2}{2}$$

- QM: first-order energy from perturbation theory

$$E_{\text{so}} = \langle n \ l \ m_l \ m_s | \underbrace{\left(\frac{e^2}{8\pi\epsilon_0} \frac{1}{m^2 c^2 \hat{r}^3} \frac{\hat{J}^2 - \hat{L}^2 - \hat{S}^2}{2} \right)}_{\hat{H}_{\text{so}}} | n \ l \ m_l \ m_s \rangle$$

Right Quantum Numbers

~~$$E_{\text{so}} = \langle n \ l \ m_l \ m_s | \left(\frac{e^2}{8\pi\epsilon_0 m^2 c^2 \hat{r}^3} \frac{\hat{J}^2 - \hat{L}^2 - \hat{S}^2}{2} \right) | n \ l \ m_l \ m_s \rangle$$~~

$$\begin{aligned}
 E_{\text{so}} &= \langle n \ l \ j \ m_j | \left(\frac{e^2}{8\pi\epsilon_0 m^2 c^2 \hat{r}^3} \frac{\hat{J}^2 - \hat{L}^2 - \hat{S}^2}{2} \right) | n \ l \ j \ m_j \rangle \\
 &= \langle n \ l \ j \ m_j | \left(\frac{e^2}{8\pi\epsilon_0 m^2 c^2 \hat{r}^3} \frac{j(j+1) - l(l+1) - s(s+1)}{2} \right) | n \ l \ j \ m_j \rangle \\
 &= \frac{e^2}{8\pi\epsilon_0 m^2 c^2} \frac{j(j+1) - l(l+1) - 3/4}{2} \langle n \ l \ j \ m_j | \frac{1}{\hat{r}^3} | n \ l \ j \ m_j \rangle \\
 &= \frac{(E_n)^2}{mc^2} \frac{n[j(j+1) - l(l+1) - 3/4]}{l(l+1/2)(l+1)}
 \end{aligned}$$

Fine Structure of Hydrogen

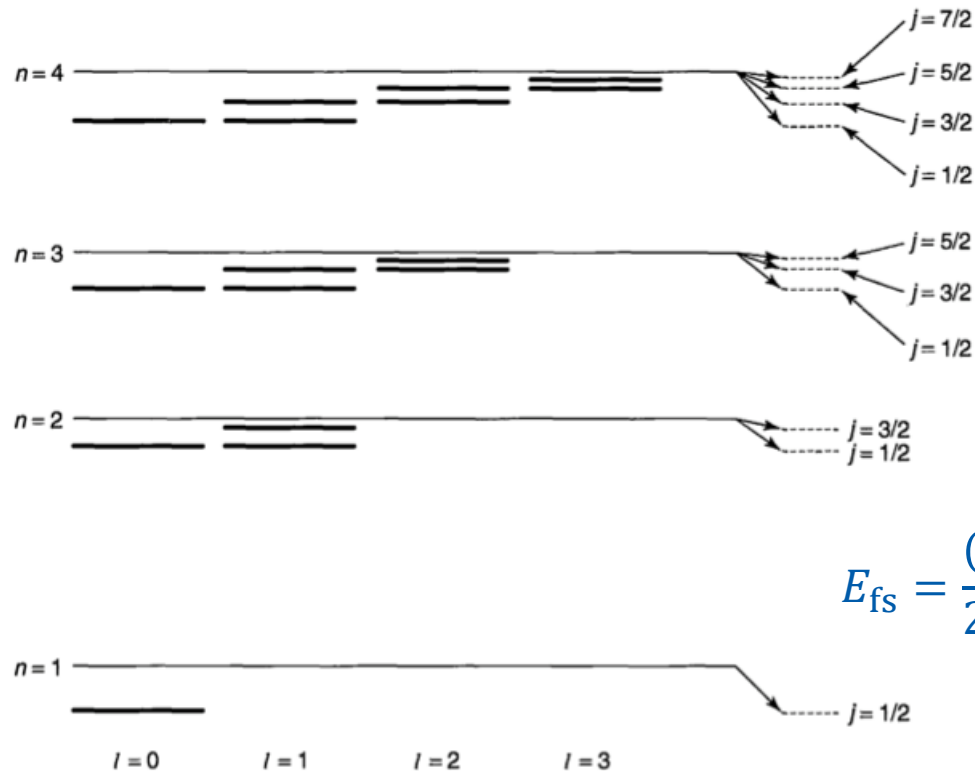
The relativistic correction and spin-orbit coupling are of the same order

$$E_{\text{rel}} = -\frac{(E_n)^2}{2mc^2} \left(\frac{4n}{l + 1/2} - 3 \right)$$

$$E_{\text{so}} = \frac{(E_n)^2}{mc^2} \left\{ \frac{n[j(j + 1) - l(l + 1) - 3/4]}{l(l + 1/2)(l + 1)} \right\}$$

$$E_{\text{fs}} = E_{\text{rel}} + E_{\text{so}} = \frac{(E_n)^2}{2mc^2} \left(3 - \frac{4n}{j + 1/2} \right)$$

Fine Structure of Hydrogen



$$E_{\text{fs}} = \frac{(E_n)^2}{2mc^2} \left(3 - \frac{4n}{j + 1/2} \right)$$

(Image: Griffiths, *Introduction to Quantum Mechanics*, 2nd Ed., Fig. 6.9)

“Good” Quantum Numbers

- 2 sets of quantum numbers: (n, l, m_l, m_s) and (n, l, j, m_j)
- In the presence of the spin-orbit coupling, the quantum states can only be well represented by (n, l, j, m_j)

Atomic Term Symbol

$$^{2s+1}[l]_j$$

- Symbols for $[l]$

l	0	1	2	3	4	...
$[l]$	S	P	D	F	G	

- Examples

- $l = 0$ and $j = 1/2$: $^2S_{1/2}$
- $l = 1$ and $j = 3/2$: $^2P_{3/2}$

Spectroscopy: Selection Rules

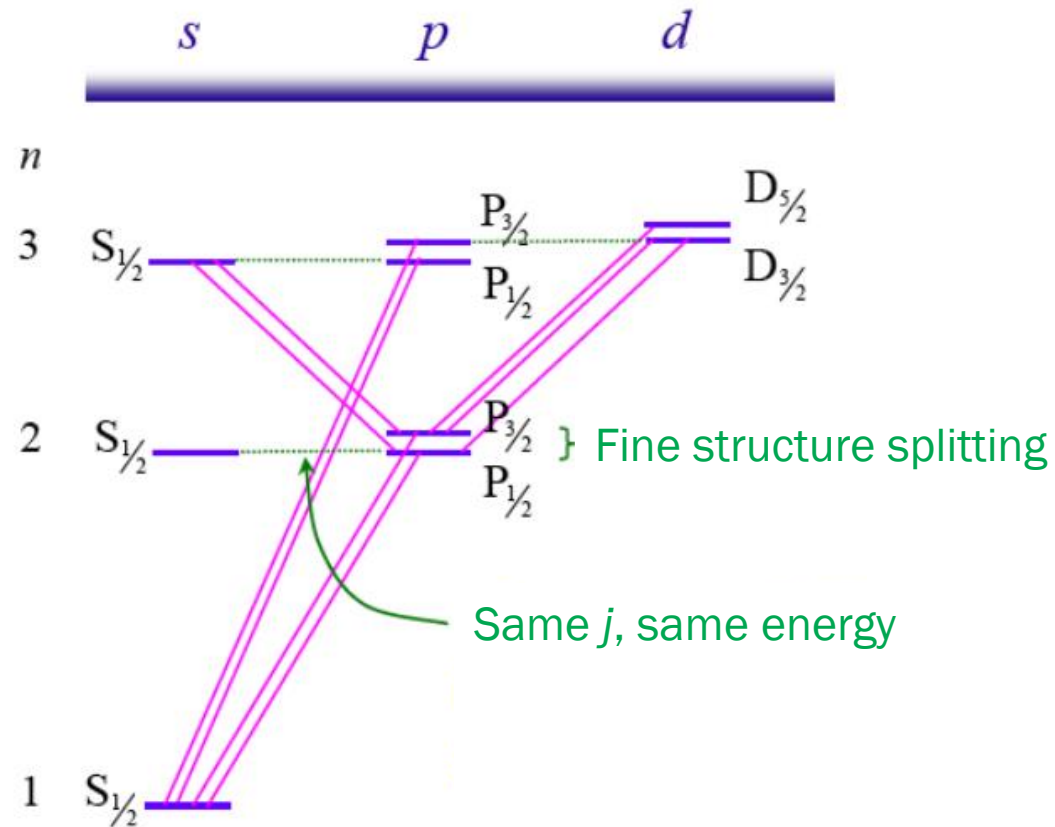
- Selection rules for a hydrogen atom

$$\Delta l = \pm 1$$

$$\Delta j = 0, \pm 1$$

- Examples
 - $^2P_{1/2} \rightarrow ^2S_{1/2}$: allowed
 - $^2S_{1/2} \rightarrow ^2S_{1/2}$: prohibited
 - $^2P_{1/2} \rightarrow ^2D_{5/2}$: prohibited

Why Two Lines?



(Image: https://nanopdf.com/download/lecture-7-fine-and-hyperfine-structure-of-one_pdf)

“Precise” Weak-Field Splitting

The energy splitting due to a weak magnetic field is

$$E_{\text{magnet}} = g(j, l)\beta_B m_j B$$

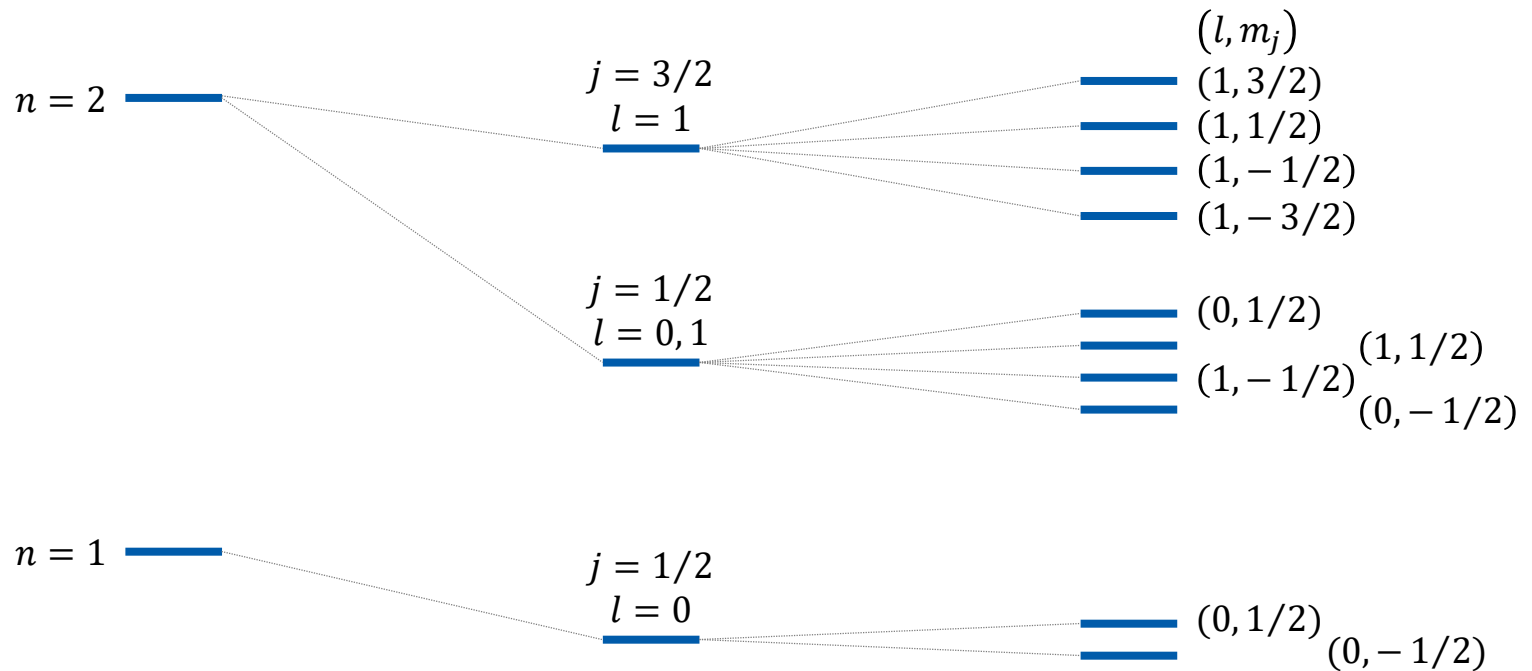
where $g(j, l)$ is the Landé g factor.

Considering the spin-orbit coupling, the energy levels are

$$E = -Z \frac{R_H}{n^2} + \frac{(E_n)^2}{2mc^2} \left(3 - \frac{4n}{j + 1/2} \right) + g(j, l)\beta_B m_j B$$

Weak-Field Energy Levels

$$E = -Z \frac{R_H}{n^2} + \frac{(E_n)^2}{2mc^2} \left(3 - \frac{4n}{j + 1/2} \right) + g(j, l) \beta_B m_j B$$



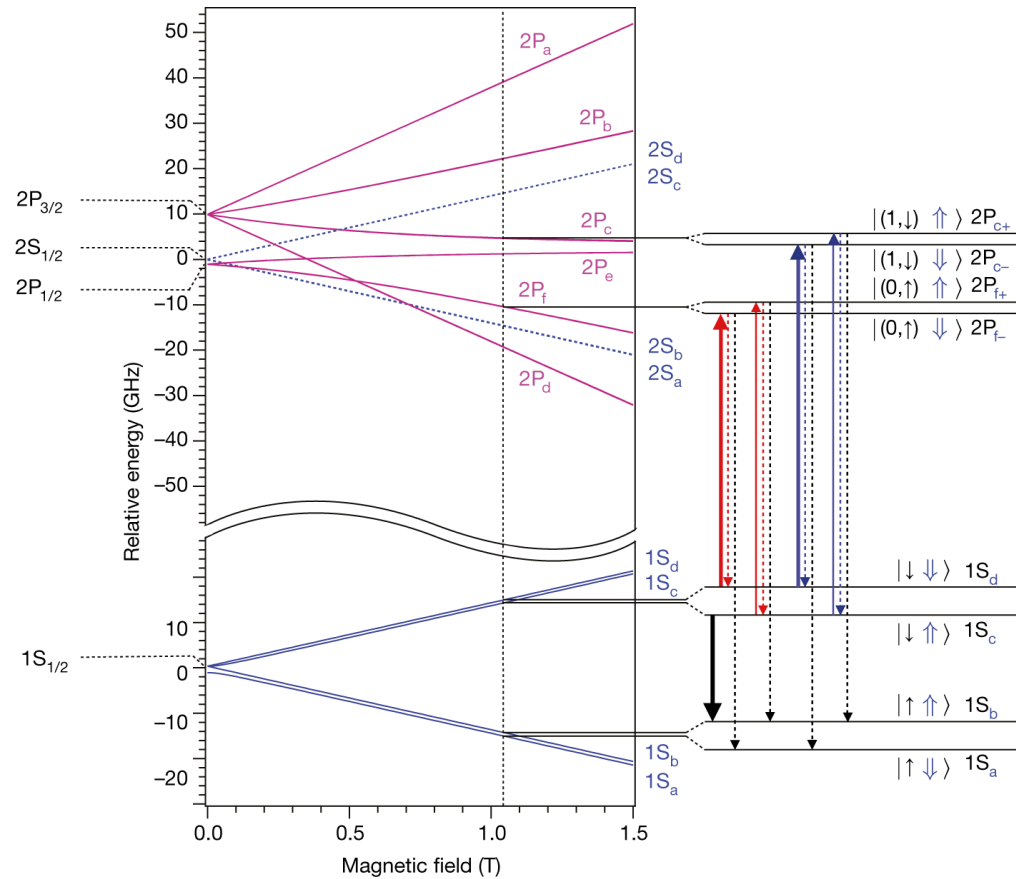
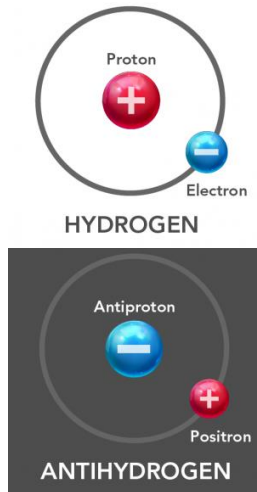
Bohr model

Fine structure

Magnetic field

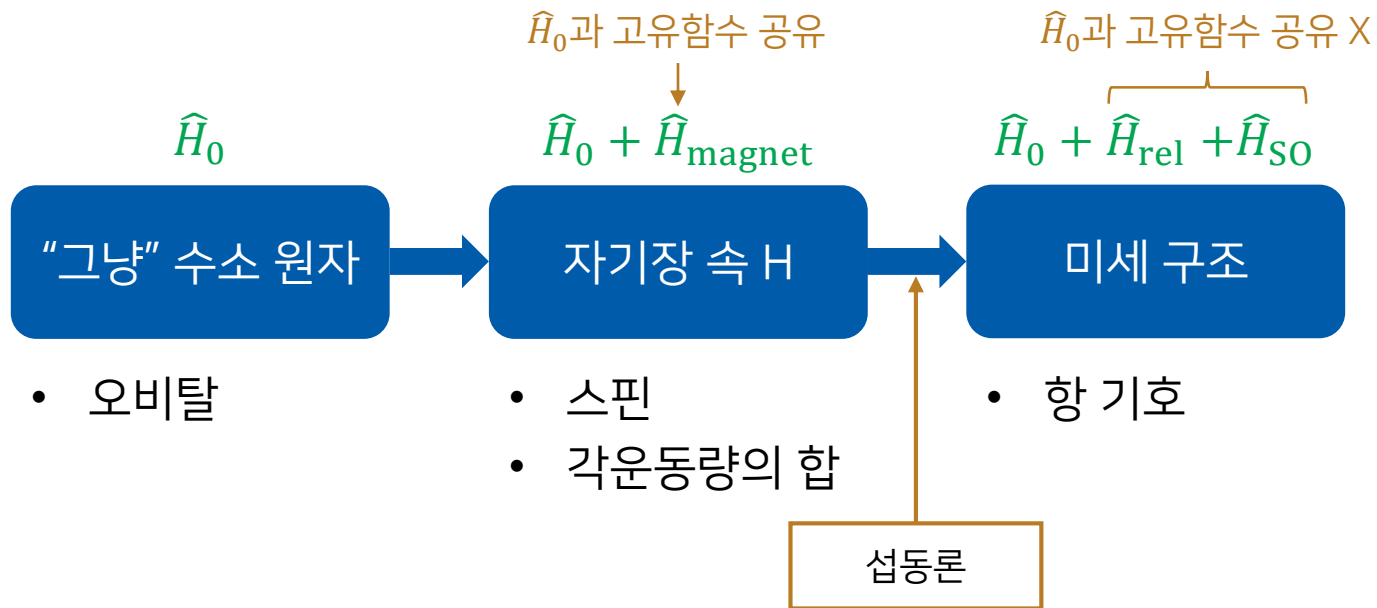
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Case of Antihydrogen



(Image: The ALPHA Collaboration, *Nature* 2020)

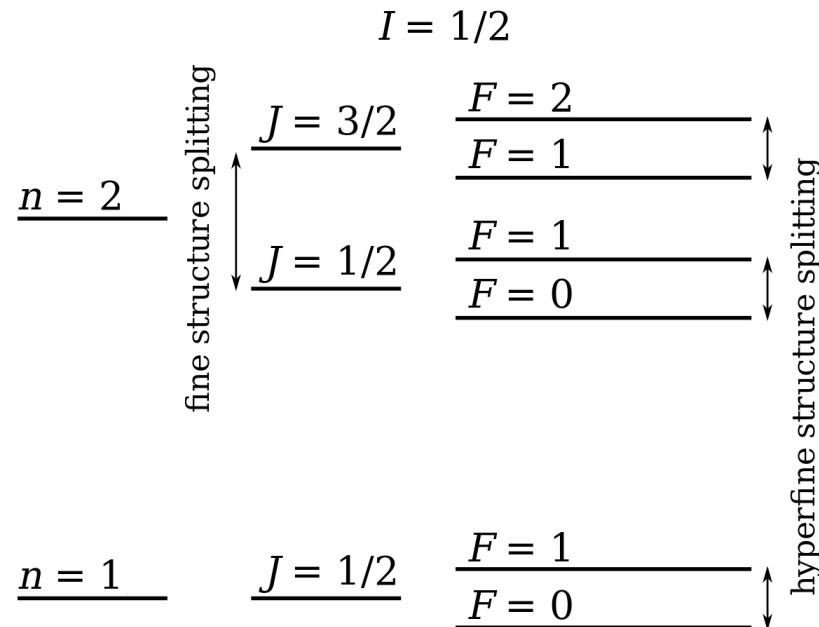
Where Are We: Hydrogen Atom



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Hyperfine Structure of Hydrogen

Due to the coupling between **nuclear spin** and **electron spin**

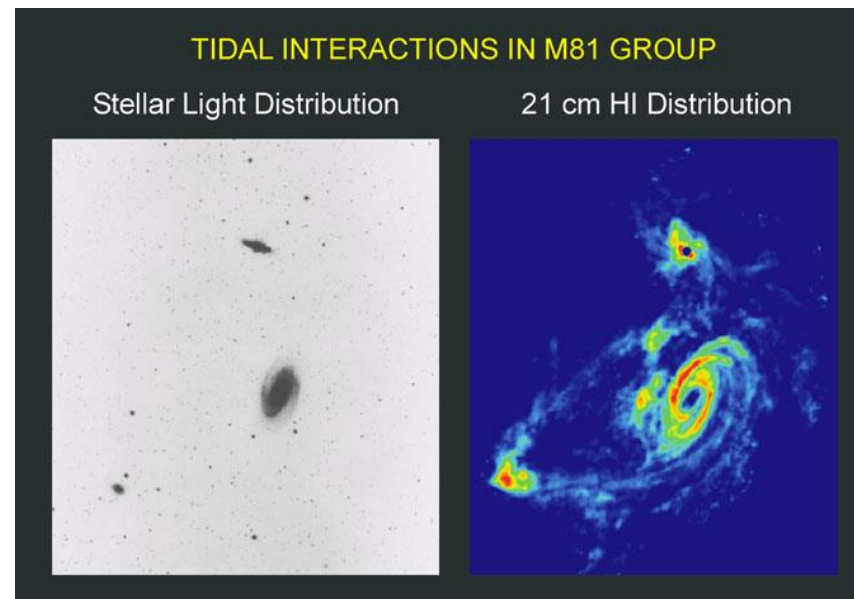


(Image: https://commons.wikimedia.org/wiki/File:Fine_hyperfine_levels.svg)

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21-centimeter Line

$$\frac{n = 1}{\quad} \quad \frac{J = 1/2}{\quad} \quad \frac{F = 1}{F = 0} \quad \updownarrow \quad \lambda = 21 \text{ cm}$$



(Image: <https://ned.ipac.caltech.edu/level5/Sept10/Turner/Turner6.html>)